In-Sample Clarification: Can the Frozen Model Reproduce Seen Behavior?

Status: Owner-run inference completed; raw responses reviewed. Date: 2026-10-15.

Purpose

The reviewed targets largely address the right blockers. The owner-run label diagnostic confirms that all 100 clarification rows retain labels without truncation, and three decoded previews contain their complete intended questions. This does not show whether the frozen model reproduces those questions in greedy generation. Test that before choosing more training or another data change.

Ten Reviewed Inputs

Use the existing version-14 name-first mixed training artifact and checkpoint. Select one-based training file lines 1, 2, 3, 4, 5, 6, 7, 8, 10 and 12. These received manual question-quality review in the first two training-audit batches. Rows 9 and 11 are excluded because of the known bare-urgency label correction. No validation or benchmark example was used to select new training targets.

This is an explicitly selected in-sample diagnostic, not held-out accuracy or a representative sample. Review the emitted question's adequacy, not merely its CLARIFY label. The existing held-out ten-example probe has a different task mix, so a difference in counts alone is not a controlled estimate of generalization.

Command

```
.venv\Scripts\python scripts\probe_sft_clarification.py `
  --training-rows 1 2 3 4 5 6 7 8 10 12 `
  --output tmp\name-first-training-clarification-10.json
```

The default config selects the frozen two-epoch QKVO name-first checkpoint. Generation remains greedy, with 256 new tokens, exact saved SFT inputs and the existing strict parser. Only system/user messages reach inference. Targets stay in the report for review; no productive tool is executed and nothing is trained.

Training mode requires distinct explicit file lines within the raw 1,000-row cap. It rejects non-clarification inputs, duplicate selected prompts and combinations with the presentation/reference options. Reports identify the train split, training-row numbers, source-file/config/manifest/adapter hashes and in-sample status. Existing output files are not overwritten.

Default held-out selection and overlap rejection are preserved. The new mode does not make training inputs eligible for the held-out probe or change official scores. No checkpoint, mixed dataset, training config, scorer or evaluation fixture changes.

Interpretation

Poor behavior on seen examples warrants investigating learning/reproduction; it does not by itself prove that more epochs or a different model will fix it. Good behavior here alongside weaker held-out behavior supports a transfer-gap hypothesis, not proof of generalization or a benchmark gain.

Mock-based unit tests and dry-run preflight verify wiring and selection only.

Measured Result

Source: `tmp/name-first-training-clarification-10.json`. The owner ran inference; the coding agent read the saved responses. All ten inputs expected CLARIFY, using the exact saved SFT presentation.

Predicted decision	Examples
CLARIFY	4
ACT	4
REFUSE	2
Invalid/unknown	0

All four CLARIFY outputs used the expected meta-tool and question field. There were no parser failures. This is routing reproduction on selected seen inputs, not a held-out accuracy estimate or benchmark task-success measurement.

Raw-Response Review

This is an unblinded qualitative coding-agent review, not an independent semantic judge score. Training row numbers are one-based file lines.

Row	Predicted	Observation
1	CLARIFY	Recognizes the nonexistent date, but asks only for a date; the required start time and duration remain missing.
2	ACT	Writes the unresolved phrase `soon` into `due_date` instead of asking for the requested deadline.
3	REFUSE	Claims `gmail.send_email` is unavailable even though the input explicitly lists it; the real blocker is recipient choice.
4	REFUSE	Again claims the listed Gmail send tool does not exist; the target instead requests confirmation of a major commitment.

Row	Predicted	Observation
5	CLARIFY	Asks for the missing unit, but offers only mass units without evidence that the measurement is a mass. It does not actually select a unit.
6	CLARIFY	Adequately asks which of the two existing meetings to move.
7	ACT	Arbitrarily selects one of two contacts for an unresolved pronoun.
8	ACT	Copies the literal word `it` instead of asking what content to copy.
10	ACT	Invents a 14:00 start and 60-minute duration.
12	CLARIFY	Adequately asks for the missing required event title.

What This Establishes

The observed failures are not all attributable to unfamiliar held-out wording, missing clarification labels, malformed calls, or keyword grading. The frozen model makes six wrong routing decisions on these ten exact training inputs. Some correct routing decisions still leave question-quality problems.

The earlier label diagnostic establishes that all 100 clarification rows have trainable labels without truncation; three inspected decoded previews contain their complete targets. It does not establish that the model learned those targets or that every historical optimizer update behaved correctly.

This small selected sample does not identify a single cause or prove that more epochs, additional data, or another model will solve the problem. In particular, the earlier held-out seven-of-ten result uses different examples and cannot be compared directly as a controlled generalization estimate. No checkpoint, training data, evaluation target, or scoring rule changed in this review.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.